

Bhumika

PhD Senior Research Scholar, IIT Jodhpur

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Research Interests

Spatio-temporal Modeling, Federated and Graph learning, On Device AI, Pervasive Computing, AI for Social Good

Research/Industry Experience

- Feb - March 2024 **University of Grenoble Alpes, France**
Visiting Researcher
- Project: Distributed Spatio-temporal Learning in Urban Areas
 - Supervisor: Prof. Phillipe Lalanda
- Jul 2012 – Jul 2013 **Samsung India Electronics Pvt. Ltd. (SEL Division), Noida (India)**
Associate Engineer
- Responsibilities: Quality Assurance of Mobile Phone, Black Box Testing
 - Manager: Er. Antim Kumar and Er. Nancy Gupta

Education

- Jan 2020 – Present **Indian Institute of Technology Jodhpur, Rajasthan, India**
Senior PhD Student, Department of Computer Science and Engineering, 8.7/10 CGPA
- Thesis Title: Spatio-Temporal Anomaly Analytics in Cityspaces with Dr. Debasis Das
- Jul 2017 – Jun 2019 **Centre for Development of Advance Computing, Noida, Uttar Pradesh, India**
M.Tech. in Computer Science and Engineering, 9.02/10 CGPA (Ranked 1st in class)
- Dissertation: Automatic Short Answer Grading with Dr. Kalpana Johari
 - University: Guru Gobind Singh Indraprastha University - Delhi (India)
- Oct 2013 – Sep 2016 **Atal Bihari Vajpayee Government Institute of Engg. & Technology, Himachal Pradesh, India**
B.Tech. in Computer Science and Engineering, 77.9% (Distinction)
- Major Project: College Bulletin Board with Mr. Ravi Kumar
 - University: Himachal Pradesh Technical University
- Oct 2009 – Feb 2012 **Government Polytechnic Sundernagar, Himachal Pradesh, India**
Diploma in Computer Engineering, 77.2% (Distinction)
- Major Project: Web Portal for Online Counseling
 - University: Himachal Pradesh Takniki Shiksha Board - Dharamshala
 - Overall Percentage:
- Dec 2008 **Matriculation, 87% (Received Merit Certificate for holding 58th Position in State)**
Major Subjects: Mathematics, Science, Information Technology, Hindi, English, Social Science,
- School: Saraswati Vidya Mandir High School Dalash - Kullu, Himachal Pradesh.
 - Board: Himachal Pradesh Board of School Education Dharamshal

Teaching Experience

- Jan 2020 – Present **Indian Institute of Technology Jodhpur, Rajasthan**
Teaching Assistant, Department of Computer Science and Engineering
- Classes: Security and Applications (Undergraduate + Postgraduate) [Outstanding TA]

Award], Edge and Fog Computing (Postgraduate), Blockchain and its Applications (Postgraduate) [Outstanding TA award], Distributed Databases (Postgraduate), Algorithms and Data Structures (Undergraduate), Artificial Intelligence (Postgraduate)

July 2019 – Dec 2019 **National Institute of Technology (NIT), Hamirpur (H.P.)**

Guest Faculty, Department of Computer Science and Engineering

- Classes: Game Theory, Wireless Sensor Network (Dual Degree)
- Lab: Computer Graphics (Undergraduate)

Jan 2019 – May 2019 **Centre for Development of Advanced Computing, Noida, Uttar Pradesh, India**

Teaching Assistant, Department of Computer Science and Engineering

Lab: Computer Graphics (Postgraduate)

Publications

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| 2024 | 1. Sandip Kumar, Pragati Sinha, Bhumika , and Debasis Das "Co-Move: Covid-19 and Inter-region Human Mobility Analysis and Prediction", IEEE Transactions on Computational Social Systems (TCSS), 2024. <i>Impact Factor: 4.5</i> |
| 2023 | 2. Nishit Bhardwaj, Anupriya Pal, Bhumika , and Debasis Das "Adaptive Context based Road Accident Risk Prediction using Spatio-temporal Deep Learning", IEEE Transactions on Artificial Intelligence (TAI), 2023. <i>Cite score: 7.7</i> |
| | 3. Vyas, Jayant, Bhumika , Debasis Das, and Santanu Chaudhury. "Federated learning based driver recommendation for next generation transportation system." Expert Systems with Applications, 2023. <i>Impact Factor: 8.665</i> |
| 2022 | 4. Jayant Vyas, Bhumika , Debasis Das, and Sajal K. Das. "Dr. MTL: Driver Recommendation using Federated Multi-Task Learning". In IEEE 98th Vehicular Technology Conference (VTC2023-Fall), Hong Kong. <i>Core Ranking: B</i> |
| | 5. Bhumika , and Debasis Das. "Deep Learning Based Urban Anomaly Prediction from Spatiotemporal Data." In Joint European Conference on Machine Learning and Knowledge Discovery in Databases, pp. 242-257. Cham: Springer International Publishing, 2022. <i>Core Ranking: A</i> |
| | 6. Bhumika , and Debasis Das. "MARRS: A Framework for multi-objective risk-aware route recommendation using Multitask-Transformer." In Proceedings of the 16th ACM Conference on Recommender Systems, pp. 360-368. 2022. <i>Core Ranking: A</i> |
| | 7. Bhumika , and Debasis Das. "UApredictor: Urban Anomaly Prediction from Spatial-Temporal Data using Graph Transformer Neural Network." In 2022 International Joint Conference on Neural Networks (IJCNN), pp. 1-8. IEEE, 2022. <i>Core Ranking: B</i> |
| | 8. Bhumika , Debasis Das, and Sajal K. Das. "RsSafe: Personalized Driver Behavior Prediction for Safe Driving." In 2022 International Joint Conference on Neural Networks (IJCNN), pp. 1-8. IEEE, 2022. <i>Core Ranking: B</i> |
| | 9. Bhumika , Jayant Vyas, and Debasis Das. "Stress-Aware Recommendation for Safe Driving using MTL-ConvLSTM." In 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC), pp. 3490-3495. IEEE, 2022. |
| | 10. Rigoni Giulio, Cristina M. Pinotti, Bhumika , Debasis Das, and Sajal K. Das. "Delivery with UAVs: a simulated dataset via ATS". In 2022 IEEE 95th Vehicular Technology Conference (VTC2022-Spring) <i>Core Ranking: B</i> |
| | 11. Vyas, Jayant, Nishit Bhardwaj, Bhumika and Debasis Das. "TransDBC: Transformer for Multivariate Time-Series based Driver Behavior Classification." In 2022 International Joint Conference on Neural Networks (IJCNN), pp. 1-8. IEEE, 2022. <i>Core Ranking: B</i> |
| | 12. Ankur Nahar, Lokendra Vishwakarma, Bhumika , Debasis Das "MetoidS: Hybrid K-Medoids-Meta Heuristic Clustering-Based Routing Optimization in Vehicular Ad-Hoc Networks" in 2022 IEEE 95th Vehicular Technology Conference (VTC2022-Spring) <i>Core Ranking: B</i> |
| | 13. Bhumika , and Debasis Das "ANOM-DGCN: Detection of Anomalies in Dynamic Networks using Deviated Graph Convolution Network" in 2022 International Wireless Communications and Mobile Computing (IWCMC) <i>Core Ranking: B</i> |

Publications Under Submission or Preparation

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| 2025 | <ol style="list-style-type: none">1. Bhumika, Phillipe Lalanda, German Vega, Debasis Das, “Federated Perspective of Spatio-temporal Crime Prediction with Skewed and Sparse Data” (<i>Under Submission</i>)2. Bhumika, Phillipe Lalanda, Debasis Das, “Graph Structured Personalized Federated Learning for Spatio-temporal Crime Prediction with Skewed Data” (<i>Under Submission</i>)3. Bhumika, Debasis Das, “Citywide Spatio-temporal Incident Prediction with Long-Tailed Spatial Distribution” (<i>Under Submission</i>)4. Bhumika, Debasis Das, “Robus Spatiotemporal Traffic Forecasting” (<i>Under Preparation</i>)5. Shruti Sureshan, Bhumika, and Debasis Das, “Leveraging Relation Networks for Few-Shot Network Anomaly Detection” (<i>Under Submission</i>)6. Nikhil Dwivedi, Bhumika, and Debasis Das, “Earthquake Severity and Probability Prediction using Deep Multi-Output Learning” (<i>Under Submission</i>)7. Jayant Vyas, Bhumika, and Debasis Das, “Escape-Route: Dynamic Spatiotemporal Graph Convolution for Evacuation Recommendation” (<i>Under Submission</i>) |
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Patent Under Submission

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| 2024 | <ol style="list-style-type: none">1. Jayant Vyas, Bhumika, Debasis Das, and Santanu Chaudhury, "Intelligent Driving-Induced Stress Detection System", 2024. (Indian Patent Published, Application No: 202411060073) |
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Scholarships & Grants

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| Nov 2020 – Dec 2024 | Ministry of Education (MoE), Govt of India.
<i>PhD Fellowship</i> |
| Nov 2024 | Registration Grant for attending CODS-COMAD held in IIT Jodhpur
<i>Special Registration Grant</i> |
| Feb 2024 | Registration Grant for attending PerCom
<i>Volunteer in PerCom</i> |
| Nov 2023 | ACM India Research Facilitation Grant (RFG)
<i>Extra page charges for Journal article</i> |
| July 2023 | ACM India Grad Cohort
<i>Travel Grant</i> |
| Dec 2022 | Indian Symposium on Machine Learning (IndoML) organized by IIT Gandhinagar
<i>Received Registration & Travel Grant</i> |
| Sept 2022 | ECML - Principles and Practice of Knowledge Discovery in Databases
<i>Free Ph.D. Paper Registration</i> |
| Aug 2017 – May 2019 | Ministry of Education (MoE), Govt of India.
<i>AICTE PG GRANT SCHEME (GATE Scholarship)</i> |

Achievements

1. Received **Outstanding TA award** for course “Blockchain” and "Security and its Applications" from the Department of Computer Science & Engineering (2023, 2022).
2. Selected as **Notable Participants** in national level IITB-FOSSEE Mapathon 2023.
3. Selected for attending the **summer school** in Climate Change AI Summer School 2023. June 23-July 27, 2023 (Virtual Mode)
4. Selected to present paper at ECML-PkDD 2022 in the **PhD Forum**.

5. Project **proposal accepted in Smart Cities Innovation Competition, (SCIC)** supported by Smart Cities Innovation Competition Committee, International Society for Urban Informatics (ISUI), 2022
6. Selected for **Research Week with Google**, organized by Google Research India, from Feb 8 to Feb 11, 2022
7. **Winner of Student's Pitch Competition** held as part of Industry Day 2021 powered by IIT Jodhpur.
8. **Second Position in Tech Ideation Competition** "Idea for India" on the occasion of Independence Day 2021, organized by Board of Co-Curricular Affairs IIT Jodhpur for our idea "Safarnama Beyond the shortest route"
9. Selected to attend **Graduate Symposium**, organized by Google Research India in the Asia-Pacific region, from April 7 to April 10, 2021
10. Research article selected for **publication in TechScape** entitled "Personalized Driver Behavior Prediction for Safe Driving using Smartphone". The Science, Technology and Education Journal of IIT Jodhpur. Vol.1, Issue 2, December 2020.
11. Qualified **GATE** (Graduate Aptitude Test) 2017 and 2018, **UGC NET** (University Grants Commission National Eligibility Test) 2018 and 2019, **HPSET** (H.P. State Eligibility Test) 2019 - Computer Science and Application

Coursework

Offline: Machine Learning-1, Natural Language Processing, Vehicular Ad-Hoc Network, Security & applications

Online: Deep Learning, Urban Computing, Machine Learning with Graphs

Academic Services

Reviewer

1. Journals: IEEE Transactions on Artificial Intelligence, Computer Networks Elsevier, IEEE TCSS
2. Conferences: IEEE ITSC, 2021, 2022, 2023, 2024, 2025

Organization

1. Organized **Hour of Code** for School children under ACM Student Chapter IIT Jodhpur 2021.
2. Organized **Student Research Symposium** under ACM Student Chapter IIT Jodhpur 2022.
3. Served as a **Technical Head** in organizing Cultural cum TechFest "Genesis" 2016.

Others

1. Served as a **member (WebMaster)** of ACM Student Chapter IIT Jodhpur between Aug 2021- Aug 2022.
2. Maintained the **official Vanet lab Website** (Jan 2023-Dec 2023)
3. Served as a **class representative** in undergraduation (2014-2016)

Software

Languages: Python, C, Java

Tools & Technologies: BlueSky simulator, QGIS, Weka, NS2, Raspberry pi, LATEX, Html, CSS, Java Script

Libraries: Scikit learn, pyTorch, pandas, geopandas, matplotlib

Extracurricular

1. Attended comprehensive **online training program** on topic "Understanding Research Metrics: Essential Tools for Academic Excellence" organized by Springer Nature for ONOS member institutions, June 2025.
2. Attended **symposium** on "AI for Smart Mobility" organized by School of AI and Data Science, IIT Jodhpur 2025.
3. Attended **PhD Conclave** on "Navigating Ph.D. and Beyond" organized by IIT Jodhpur in Jan 2025.
4. Attended a workshop on **Patent Landscaping** organized by IIT Jodhpur in Sept 2023.
5. Attended **IoT and Applications** Workshop organized by IoT - IDRP, IIT JODHPUR Aug 2023.
6. Presented a **poster** at Industry Day organized by IIT Jodhpur 2023.
7. Presented a **poster** at IndoML organized by IIT Gandhinagar 2022.
8. Participated in the **Science of Self Discovery** Workshop in 2022.
9. Attended an **ACM IKDD Event Collocated With KDD2021** on title "Data Science In India" in Aug 2021.
10. Attended **ACM W India Grad Cohort Workshop** in 2020, 2021, 2023.
11. Second Position in **Group singing** in the fest "COLOSSUS-2019" held from 27th Feb- 1st March 2019.
12. Participated in **Kho-kho** in the fest "COLOSSUS-2019" held from 27th Feb- 1st March 2019.